

From gait symmetry to useful movement prediction

A guide to our research questions, findings, and next experiments. Evidence and workshop information reviewed on 7 September 2026.

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1. The question connecting the work

What should a JEPA learn about human movement, and how can we tell whether pretraining has helped?

Our experiments study sequences of estimated body landmarks, including shoulders, knees, and ankles. An encoder turns these coordinates into numerical features that a prediction model can use. During Joint-Embedding Predictive Architecture (JEPA) pretraining, part of the input is hidden, and a predictor estimates its features from the visible observations. A second, slowly updated encoder supplies the target features. This training does not use gait-condition labels.

The motivating idea is that predicting missing observations could encourage a model to represent meaningful relationships between body parts and their movement. For example, the visible hip and knee trajectory might help predict an obscured ankle. We evaluate that idea by holding the encoder's weights fixed—often called freezing the encoder—and testing whether its features help a simple regression model predict a movement quantity on videos excluded from training.

The latest result makes this question more pressing. The complete masking comparison now contains 50 trained encoders, but selecting gait-relevant landmarks as hidden targets does not establish better prediction than selecting targets across the body. Both trained alternatives also perform worse than their untrained reference on average under the current evaluation. This outcome gives us a useful research direction: investigate which parts of preparation, training, and prediction preserve access to movement information, before making a larger model or adding more training objectives.

Three questions are especially interesting now:

1. **Where does useful movement information become difficult to recover?** The recorded coordinates, prepared input, learned features, and final feature summary can each affect the result.
2. **Which left–right relationships should the representation preserve?** Overall movement and the side contributing more movement need different responses to reflection.
3. **Can learning these relationships improve prediction of future movement?** This would provide a stronger connection to temporal world models than a test of whole-clip consistency alone.

This tutorial separates measured results on gait recordings from synthetic demonstrations and proposed experiments. The new masking result is empirical evidence; the symmetry-training and forecasting extensions remain demonstrations awaiting real-data evaluation.

Section 4 examines whether our masking choices are too restrictive. It distinguishes the landmarks supplied to the encoder from those selected as hidden targets, explains established alternatives, and proposes a manageable comparison grounded in the current results.

2. The data and the research trajectory

A curated movement dataset and a controlled evaluation

The completed evaluations use 625 accepted clips from 93 source videos selected from the Gait Abnormality in Video Dataset (GAVD). GAVD is a researcher-curated collection of online gait videos, with annotations covering normal, pathological, and other abnormal walking patterns. Its YouTube sources were assembled for gait analysis, rather than sampled as arbitrary internet videos. This clinical motivation is important, although our current prediction target is calculated from estimated coordinates and is not a diagnosis or an independently measured clinical score. [GAVD dataset paper](#)

We divide source videos into five groups. For each group, every clip from its videos is excluded from both encoder training and fitting the final regression model. We repeat training with five random seeds, which change the starting weights as well as sampled clips, hidden targets, and training views. Thus, two training choices require $5 \text{ groups} \times 5 \text{ seeds} \times 2 \text{ choices} = 50$ trained encoders. The masking extension has completed this full design, separately from the original reflection evaluation.

Consider a video contributing ten clips. Testing on two of those clips while training on the other eight could allow the model to benefit from the same recording conditions or repeated movement. Keeping the entire video together prevents that overlap. When scoring, each video receives the same total weight, so a video with many clips cannot dominate simply through its clip count. This is careful recording-level separation; without verified participant identities, it cannot establish separation of people who might appear in different videos.

How one question led to the next

We began with an anatomical choice. Gait-relevant landmarks seemed sensible targets for learning from missing body observations. The initial design used that selection throughout, however, so it could not show whether the anatomical choice caused an improvement.

We then used laterality to test a known relationship. Here, laterality means a signed contrast between movement on the left and right anatomical sides. For illustration, a contrast of $+0.2$ should become -0.2 when we reflect the coordinates and exchange the left/right landmark labels. This follows from the definition of the measurement; it does not require the person's gait to be symmetric.

The reflection evaluation separated consistency from predictive value. Adding mirrored clips during training reduced the specified disagreement between original and reflected features. Its effect on movement prediction remained uncertain: the estimated R^2 improvement was 0.004, with a 95% interval from -0.006 to 0.013. Ordinary pretraining also failed to establish an improvement over the matched initial encoder: the R^2 difference was -0.018 , with an interval from -0.039 to 0.002. These intervals resample whole source videos while keeping the fitted models fixed. [Completed results notebook](#)

A sign-reversal rule imposed on the final output can satisfy the expected relationship even with an untrained encoder. That control showed why satisfying a mathematical rule cannot, by itself, establish useful learning. The feature test also assumes that reflection exchanges anatomical tokens without changing their internal feature coordinates; other possible responses within those coordinates remain untested.

The new notebooks investigate what might explain the gap. Notebook 07 follows movement information through the pipeline. Notebook 08 isolates the choice of hidden targets and now supplies a complete real-data comparison. Notebook 09 introduces explicit reflection training, and Notebook 10

asks whether future-feature prediction produces features useful for forecasting. These questions build on the earlier work without treating every new notebook as a completed contribution.

For navigation, notebooks 00–05 contain the original protocol and evaluation; 06 checks prerequisites for external evaluation; 07 provides diagnostics; 08 compares masking choices; 09 tests reflection training; and 10 introduces past-only forecasting. External participant-level validation has not yet been completed.

3. What the updated notebooks help us answer

Question 1: Does the input still express the movement we want to predict?

Notebook 07 compares the original movement measurement with the same formula applied after input preparation. Preparation fills eligible short gaps, normalizes coordinates, and resizes sequences to a common length. These operations are useful for batching, but they can also change a quantity based on movement between observations.

In a read-only reconstruction, the two calculations have matching signs in about 70% of cases, with equal total weight per source video. Both calculations are finite for 623 clips from 92 sources; two clips have no finite recomputation. This measures agreement between two ways of calculating the target. It is neither a model's accuracy nor a ceiling on what a model could learn.

A simple example explains a separate concern about summarizing time. Let a left landmark follow positions (0, 1, 0, 1), and a right landmark follow (0, 0, 1, 1), at equally spaced times. Their average positions are both 0.5, but their median movement speeds are 1 and 0. Swapping the trajectories reverses the speed contrast while leaving the averages unchanged. A model given only those average positions cannot distinguish the cases. Averaged encoder features could still contain motion if the encoder has represented it before averaging.

The next useful comparison is to apply the measurement after each preparation step on the same clips, then compare time-averaged features with summaries that retain ordered movement. If a change restores measurement agreement and improves held-out prediction, it would identify a practical weakness. If agreement improves without better prediction, we would need to investigate representation learning or the regression model as well. The existing diagnostic motivates these tests but does not identify which operation causes the discrepancy.

Question 2: Does hiding gait-relevant landmarks help learning?

Notebook 08 compares hidden targets selected from twelve landmarks—the left and right shoulders, hips, knees, ankles, heels, and foot tips—with targets selected across all 33 landmarks. A target token represents one landmark over four time steps.

The important control is the number of valid targets actually hidden. Suppose there are four time blocks and every landmark is visible. Twelve landmarks provide 48 candidate tokens, while 33 provide 132. Hiding half would produce 24 targets in one condition and 66 in the other. The notebook instead hides the same number in both conditions, adjusting the shared count when observations are missing. This avoids attributing a change in training difficulty merely to hiding more input.

The paired models share starting weights, source-video draws, training views, and 1,200 optimizer updates. Both also use the same gait-landmark pooling in their feature-variation regularizer, which discourages uninformative representations. Consequently, the comparison tests **where masked prediction targets are chosen within an already anatomy-informed training procedure**. It does not remove every anatomical influence from the uniform-target condition.

The full five-group, five-seed grid is now complete. The table below was recomputed from the retained

test predictions during this review; no new training was performed. “Direct pose summaries” means average landmark positions, their variation over time, average movement between prepared samples, and the fraction of observations available for each landmark. This baseline uses the same prepared input without an encoder.

Features supplied to the regression model	R^2	Mean absolute error
Encoder pretrained with gait targets	−0.540	0.0564
Encoder pretrained with targets across the body	−0.423	0.0562
Matched encoder without pretraining	−0.351	0.0526
Direct pose summaries	−0.004	0.0449
Mean target from training videos	−0.011	0.0462

For each seed, predictions are combined across the five held-out groups before scoring, with equal total weight per video. The table then averages the five seed-specific scores. Direct pose and training-mean predictions do not depend on the training seed and are not five independent baseline runs.

R^2 compares squared prediction errors with variation in the observed target. A value of zero matches the source-weighted mean of the evaluated targets; a negative value has larger squared error. That evaluation mean is a mathematical reference, not a deployable predictor with access to test labels. The last table row is the usable control fitted from training videos only. Mean absolute error measures the average size of the error in units of the normalized movement contrast; smaller values are better, and these values are not percentages.

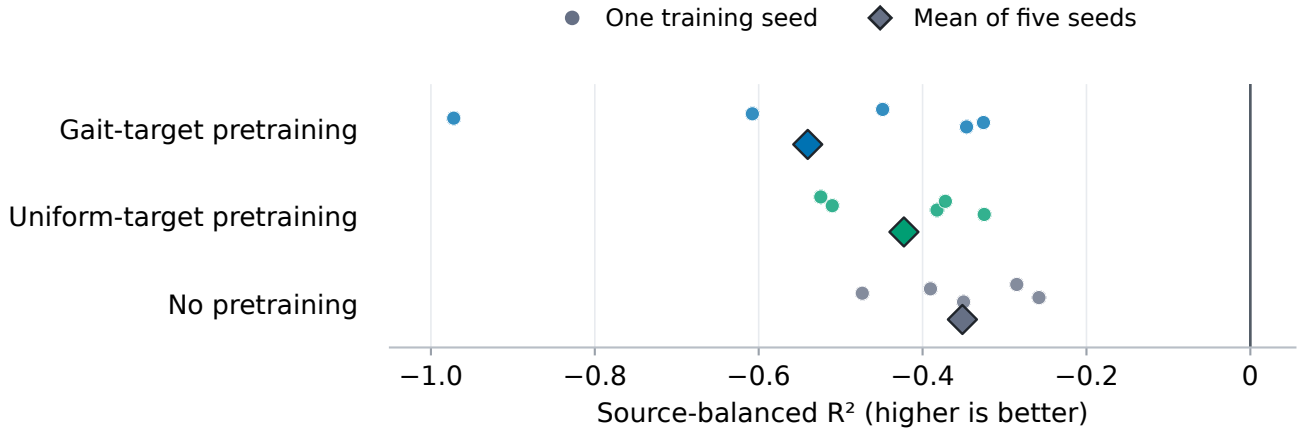


Figure 1. Prediction from the frozen encoder in the completed masking comparison. Small circles show the five seed-specific R^2 scores; the larger diamond marks their mean. Scores farther right are better. The spread describes sensitivity to random training choices, not a confidence interval; small vertical offsets keep neighboring marks visible. All three encoder conditions have negative R^2 for every seed under the current regression procedure.

Selecting gait targets improves R^2 for two seeds and worsens it for three. Its average R^2 difference relative to targets across the body is −0.117. An exploratory 95% paired source-bootstrap interval, calculated during this review from the saved predictions, runs from −0.290 to 0.016. The interval keeps the existing models fixed and does not include uncertainty from retraining. It permits a small benefit as well as substantial harm under this procedure, so the experiment does not establish an advantage or precise equivalence between the choices.

The nearly identical mean absolute errors add useful context: the large R^2 difference does not describe

a uniform increase in error on every clip. Because squared error gives extra weight to large mistakes, inspecting which videos and training runs produce those mistakes is more informative than assuming one broad anatomical explanation. Difficult cases must remain in the evaluation unless an independently specified eligibility rule excludes them.

Both pretrained encoders have higher mean absolute error than the initial encoder for all five seeds. Training loss nevertheless decreases over the 1,200 updates. Optimization has therefore progressed without establishing an improvement in this prediction task. This finding concerns the tested training and regression procedure; it cannot locate the cause or show that all movement information is absent.

One particularly useful follow-up concerns the regression model. Encoder features have 960 dimensions, compared with 264 for direct pose summaries, and the extension uses a fixed ridge penalty of 1. This penalty limits the size of the fitted regression weights to reduce overfitting. The same value can constrain differently structured features by different amounts. Selecting it using training videos only, with the same selection procedure for every representation, is an important diagnostic before treating poor prediction as a property of the encoder alone.

The absolute scores also should not be compared with the original reflection study as if only masking changed. The extension uses a different learning-rate schedule, teacher update schedule, mask fraction, and regression-selection procedure. The within-extension masking comparison is controlled; the numerical gap between evaluations has several possible causes.

Question 3: Can explicit reflection training preserve information about each side?

Notebook 09 compares the base objective, mirrored training clips, and an explicit penalty for disagreement between corresponding original and reflected landmark features. The penalty encourages agreement after anatomical landmark exchange, with missing observations handled consistently.

Imagine left and right movement values of 3 and 2. Their average remains 2.5 after exchanging sides, while their difference changes from +1 to -1. A useful representation may need to support both quantities. In plain language, some information should stay the same under reflection, while information about which side contributes more should change predictably.

Perfect feature agreement is insufficient evidence of success: giving every clip the same feature vector would also make the comparison agree. The notebook therefore measures predictive performance and checks whether features vary across clips. These checks can expose a trivial solution, although feature variation alone does not establish useful movement content.

The retained run uses generated data and eight training updates. The explicit penalty slightly reduces its targeted reflection error without a displayed predictive advantage; there is no real-data result establishing that it helps gait learning. Its short run is a software and teaching demonstration, not a test of an adequately trained method. The extra reflection calculations also cost computation, so matching update counts alone would not support an efficiency claim.

A useful real-data outcome would be better movement prediction together with better geometric agreement. If only agreement improves, the contribution remains a consistency result. If prediction deteriorates, the next test should examine where the penalty is applied and how strongly, rather than assume that more symmetry training is beneficial.

Question 4: Does learning future features help predict future movement?

Notebook 10 observes the first 0.8 seconds of a sequence and predicts landmark positions 0.25, 0.50, or 0.75 seconds beyond that boundary. For example, the middle horizon asks about an ankle around 1.3 seconds, using observations available through 0.8 seconds. Unlike the masked-learning notebooks, this

forecasting implementation supplies only the twelve selected landmarks to its past and future encoders. Whether a broader input helps is a separate, untested comparison.

This changes the information available to the model. A whole-clip movement score can use the complete sequence; forecasting must prepare the input from the observed past alone. The notebook preserves timestamps and calculates its reference position and body scale using only that prefix. A test changes future coordinates and visibility while checking that the prepared input and prediction stay unchanged.

During pretraining, a predictor estimates features of a future window. During evaluation, a separate regression model maps the frozen encoder's past features to future coordinates. The present notebook therefore tests whether future-feature training improves the **representation of the observed past**. It does not yet decode the JEPA predictor's future feature vector or evaluate a sequence of predicted future states.

The comparisons include keeping each landmark at its last position, continuing its last estimated velocity, regression directly from past coordinates, the initial encoder, and training against a future taken from a different training video. That mismatched-future condition is important: if it performs similarly, correct temporal pairing has not demonstrated added value.

The retained training results use synthetic oscillating trajectories and do not show a consistent advantage over direct past-coordinate regression or the initial encoder. Correctly paired and mismatched futures also perform similarly in that demonstration. Preparing the real recordings is feasible: a read-only check found 611 clips eligible for at least one horizon, from 93 videos, contributing 1,814 clip-horizon examples. Duration and landmark visibility affect which outcomes can be scored. These are repeated questions about existing clips, not new independent samples. Real-data forecasting performance remains unestablished.

4. Is our fixed landmark selection too restrictive?

It is a reasonable starting prior, but too narrow a basis for concluding which masking strategy helps movement learning. The twelve-landmark choice has anatomical motivation; our experiments have not established that this exact set is neurologically optimal. “Gait-focused” or “anatomically motivated” describes the evidence more accurately than “neurologically validated.”

The concern also needs a precise account of what is fixed. A fixed number of hidden observations can make a comparison fair. Permanently restricting which landmarks can become targets limits the questions that comparison can answer.

What is—and is not—restricted in the current notebooks

In notebooks 03, 08, and 09, the encoder receives all 33 landmarks. The original recipe selects hidden prediction targets from twelve of them, but draws a fresh set of hidden landmark-time tokens at each training update. The same anatomical list is reused; the actual missing observations change. Valid elbows, wrists, and other landmarks outside that list remain available as context, and the target encoder processes the unmasked full-body input.

There are additional anatomical choices elsewhere in the pipeline. The feature-variation regularizer summarizes the twelve gait landmarks from unmasked training views, including in Notebook 08's all-landmark masking condition. The laterality readout uses five bilateral pairs: shoulders, knees, ankles, heels, and foot tips. Hips belong to the masking set but not this readout. Notebook 10 makes a stronger restriction by supplying only twelve landmarks as input. Changing any of these choices answers a different question from changing the mask alone.

We have therefore already tested one important alternative: Notebook 08 selects the same number of hidden targets across all 33 landmarks. That complete comparison did not establish an advantage for

the gait-focused selection. Several random twelve-landmark target sets would be another useful control; the code supports a preselected random subset, but no such real-data comparison has been retained.

A numerical example reveals a second restriction. With 64 prepared time steps and four steps per token, all 33 landmarks provide $16 \times 33 = 528$ tokens. The twelve-landmark target set provides $16 \times 12 = 192$ candidates. Hiding half of those candidates hides 96 tokens, or about **18% of the full input**, assuming all observations are valid. Thus, the current “50%” setting does not hide half of the body sequence. Missing observations further limit the actual shared count.

This leaves many neighboring observations available. A short ankle gap may be predictable from the ankle immediately before and after it, without requiring a rich account of whole-body movement. That is a plausible explanation to test, rather than an established cause of our weak prediction results. In a JEPA, the targets are contextualized features, so ease of coordinate interpolation alone cannot establish how the learned predictor solves its task.

Mask shape changes the information available for prediction

The following figure holds the hidden count constant while changing its arrangement. It makes visible why selecting a landmark set, choosing a masking percentage, and choosing a pattern over time are distinct decisions.

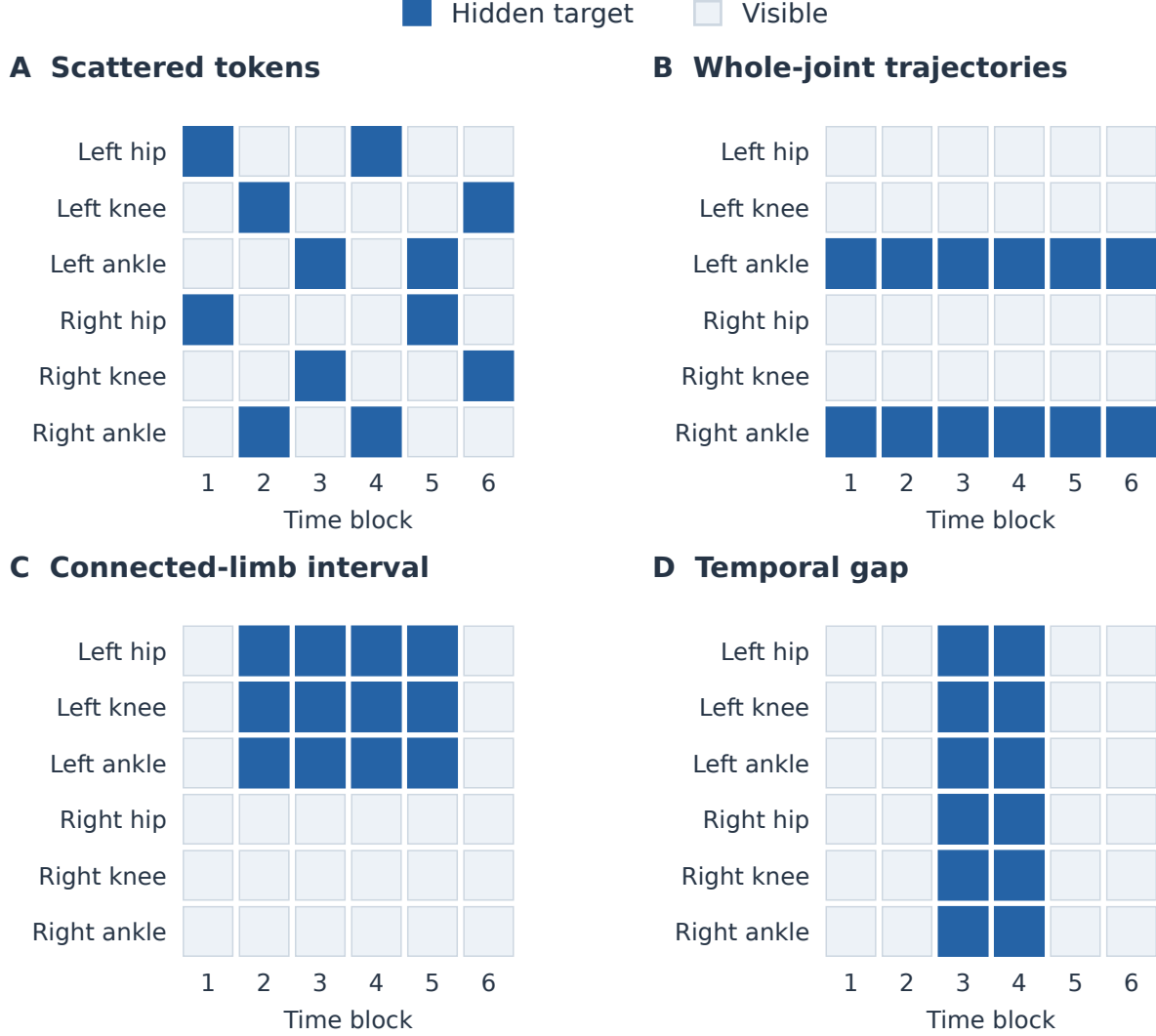


Figure 2. Four ways to hide 12 of 36 landmark–time cells. Blue cells are prediction targets; pale cells remain visible. This is a hypothetical six-landmark example, not the experimental landmark set or masking rate. Panel D illustrates completion, with observations before and after the gap.

Established and promising alternatives

The literature contains established baselines as well as newer proposals. Their evidence comes mainly from action recognition, pose reconstruction, or RGB video, so their relevance is a reason for comparison rather than a prediction that they will improve our gait endpoint.

1. Scattered random masking with broader or softer landmark coverage. Uniformly sampling joint–time tokens across the body is an important baseline, already present in Notebook 08. Random patch masking is central to MAE, whose image experiments favored it over the grid and block alternatives tested there. That result does not determine the best skeleton mask, but it shows why a simple random baseline deserves serious treatment. [MAE, CVPR 2022](#)

Two additional controls would clarify the anatomical claim. First, compare the selected gait set with several random twelve-landmark sets chosen before looking at outcomes. Second, give gait landmarks a sampling preference while keeping every valid landmark eligible. For example, an equal mixture of

all-landmark and gait-only sampling would soften the exclusion rule. That mixture is a proposed starting comparison, not an established optimum. If it outperforms the fixed set, the useful ingredient might be anatomical emphasis combined with broader coverage.

The landmark scheme itself matters: several of the 33 points describe the face or hands, so uniform sampling of points is not uniform sampling of body regions. A body-region-balanced draw would be another reasonable control, provided its grouping is defined in advance. A fixed random subset and a new subset drawn at every update should also be distinguished: the first tests a particular restricted pool, while the second can eventually cover the whole body.

2. Motion-weighted masking. Instead of always favoring the same landmarks, assign higher masking probability to joint-time regions with more observed movement. A swinging arm may receive more targets in one clip, and a moving foot in another. Selection remains stochastic, so this need not always hide the single fastest-moving joint.

This is the most direct missing comparison with the source method. The original S-JEPA uses motion-weighted masking inherited from MAMP; our fixed gait-target pool is an adaptation. MAMP's mask-only ablation reports NTU-60 cross-subject linear-evaluation accuracy of 84.9% with motion-weighted masking versus 83.7% with random masking. MAMP also changes the prediction target to motion, so its overall performance cannot be attributed to mask selection alone. We should first transfer the sampling rule while retaining our JEPA feature target. [S-JEPA, ECCV 2024](#), [MAMP, ICCV 2023](#), [Table 8](#)

For laterality, movement magnitude is an imperfect guide. A limb that moves less can be essential to the left-right contrast, while an implausibly large jump can come from a tracking error. A practical adaptation should preserve a nonzero chance of masking slow-moving regions, exclude invalid observations, and control the influence of isolated jumps. These are proposed safeguards to evaluate, not evidence that motion-weighted masking already works here.

3. Whole-joint trajectories and connected body regions. A trajectory mask hides the same landmark throughout the input window. For example, withholding the left ankle for the whole clip removes the immediate before-and-after observations that could fill a short gap. VideoMAE's analogous "tube" mask repeats a spatial mask through the video. At the same 90% masking ratio, its reported Something-Something V2 accuracy is 69.6% for tube masking and 68.3% for independent random masking. Those are fine-tuned RGB action-recognition scores; the very high ratio should not be copied directly to sparse pose input. [VideoMAE, NeurIPS 2022](#)

A connected-region mask hides several anatomically linked landmarks for a continuous interval. For example, hide the left knee, ankle, and foot together while leaving the hip, trunk, and opposite leg available. This asks whether broader body relationships help when local cues are missing. Groups should follow anatomical connections rather than consecutive indices in a data array.

There is close skeleton-specific precedent. The SLiM preprint, revised in August 2026, masks connected anatomical regions over consecutive time blocks and varies temporal duration with region size. It combines teacher-feature prediction with global/local alignment and reports structured-mask comparisons. Its design makes connected-region masking a well-motivated comparator, while ruling out a broad novelty claim based on introducing anatomical tubes alone. [SLiM, 2026 preprint](#)

I-JEPA also motivates predicting several connected targets from surrounding context, although its blocks are image regions rather than body parts. Its preference for multiblock masks, alongside MAE's preference for random masks in a different objective, argues for testing mask design within the actual learning task. Hiding an entire asymmetric limb can also make its movement genuinely ambiguous; greater difficulty does not guarantee more useful features. [I-JEPA, CVPR 2023](#)

4. Missing time intervals and strictly future targets. A temporal gap hides every landmark

during part of a sequence. With observations before and after the gap, the model performs movement completion. MotionBERT uses joint-level and frame-level corruption, while PoseBERT studies missing poses and temporal blocks. Both provide precedent for varying which parts of time are observed, although their reconstruction objectives differ from our feature-prediction objective. [MotionBERT, ICCV 2023](#), [PoseBERT](#)

Hiding the entire future asks a different question. Observing through 0.8 seconds and predicting an ankle at 1.3 seconds requires continuation without later clues. Notebook 10 implements that information boundary, but has no retained real-data training result. The August 2026 Human-JEPA preprint offers adjacent evidence from human RGB video by comparing forecasting-oriented masks with other masking choices. Its results motivate testing this choice without establishing a benefit for our skeleton task. [Human-JEPA](#)

Joint masks, middle gaps, and future masks can eventually be combined during training. First compare them separately so that a gain can be attributed to a particular source of supervision. Preparing a whole clip and hiding its future afterward can still expose future information through normalization or interpolation; the forecasting pipeline must continue to prepare its input from the observed past alone.

5. Complementary masks and several levels of difficulty. Instead of presenting two similarly incomplete views, deliberately hide different kinds of information in each. One view might retain the trunk while hiding moving limbs; another might retain those limbs and hide some trunk observations. ASMa studies complementary choices based on joint connectivity and frame motion, within a method that also changes the encoders and feature-alignment objective. It supports testing complementary coverage, rather than attributing the full method's gains to one mask. [ASMa, TMLR 2026](#)

Short gaps, long gaps, small regions, and larger regions can also be sampled with different frequencies. This gives the encoder varied prediction tasks, but broad difficulty variation should follow an interpretable fixed-pattern comparison. A schedule that makes masking harder over training would be another hypothesis; we have no result showing it is preferable here.

For this signed endpoint, caution is needed about making incomplete views identical. Removing the more active leg changes the information available about laterality. A loss that encourages agreement regardless of which limb is missing could suppress the very distinction we want to retain. Evaluate side-sensitive prediction alongside any agreement between views.

6. Learned or adaptive selection. A small auxiliary model can learn which observations to reveal or hide. AdaMAE learns a sampling distribution over the tokens that remain **visible**, using a reconstruction-error-based training signal. This differs from a rule that hides high-motion regions. A pose adaptation could learn which observations make the remaining movement predictable, but would add parameters, computation, and another source of training instability. [AdaMAE, CVPR 2023](#)

I would postpone this until simpler masks provide an informative baseline. A learned selector could prioritize pose-estimation errors because they are difficult to predict, or communicate information through the locations it chooses. Using complete examples to construct self-supervised training masks is not automatically label leakage, but a future-informed mask must never influence an input advertised as past-only. An adaptive method should also beat a simple motion-based rule after accounting for its extra cost.

Changing the target or the loss is a separate experiment

Some promising methods broaden the learning signal without changing mask placement. V-JEPA 2.1 supervises visible as well as hidden tokens and intermediate encoder layers. AMR assigns greater reconstruction-loss weight to movement-rich regions and also changes the prediction architecture. Neither should be described as merely an alternative sampler. Keep the mask fixed when testing these changes,

so a result can be attributed to what the model predicts or how its errors are weighted. [V-JEPA 2.1, 2026 preprint](#), [AMR](#), [CVPR 2026](#)

The same distinction applies to broadening the regularizer's twelve-landmark summary or Notebook 10's input. Those choices deserve investigation, but changing them together with masking would prevent a clear explanation of the result.

Which comparisons should we prioritize?

The most useful next experiment is a focused comparison of what information is hidden, rather than a large search over every available technique. The following order balances scientific value with the weakness of the current predictive results.

1. **Make the evaluation informative before expanding training.** Use training-source validation to check regression regularization and motion-sensitive feature summaries, retaining the untrained encoder and direct-pose controls. A masking comparison needs a way to reveal useful learned information if it is present.
2. **Test the anatomical exclusion rule.** Keep scattered masks and compare the gait-only and all-landmark references with several preselected random twelve-landmark sets and one soft gait preference. Judge random sets as a declared group rather than reporting whichever set gives the most favorable contrast. If a new condition changes the feasible shared token budget, repeat the reference conditions at that budget.
3. **Test motion and mask shape under all-landmark eligibility.** Prioritize motion-weighted selection because it is the missing canonical S-JEPA comparator, and connected-limb intervals because they remove nearby body cues. A whole-joint trajectory condition would distinguish missing one trajectory from missing a connected region. Ensure the body-region groups together cover the declared landmark pool; otherwise the structured mask quietly introduces another exclusion rule. Keep the feature-prediction objective and all other training choices fixed for these comparisons.
4. **Evaluate temporal prediction separately.** Complete the real-data past-only comparison, including direct past-pose regression, initial features, simple motion continuation, and mismatched futures. Compare twelve- and all-landmark inputs only as an explicitly separate question. Future prediction should be scored at the same times on the same supported observations.
5. **Add complex combinations only when simpler results justify them.** Complementary views, visible-token supervision, and learned selectors become more informative after identifying which mask types preserve useful movement. A favorable training loss alone is insufficient grounds to expand the method.

A modest training-only pilot can establish feasible settings before confirming a small declared comparison across the five video groups and five seeds. It should not select the winning recipe from the outer test results.

How to keep these comparisons fair and interpretable

Match the number of genuinely observed tokens deliberately hidden, while recording the fraction of valid input this represents. Naturally missing landmarks must remain distinct from observations deliberately withheld for prediction. A whole-joint or whole-frame mask may only support certain counts; choose feasible budgets or report a controlled range instead of silently breaking the mask's structure to reach an arbitrary total. Overlapping blocks should not inadvertently count the same target multiple times.

For example, on the fully observed 16-time-block, 33-landmark grid, a complete joint trajectory contributes 16 hidden tokens and a complete time block contributes 33. A 96-token budget accommodates six trajectories, but not an integer number of complete time blocks. These two mask families have no

common positive count short of hiding the entire input. One solution is to compare each structured mask with a scattered-mask reference matched to its actual count, and report the different budgets explicitly. Their effects relative to those references would not establish a direct ranking of the two structures at an identical budget.

Equal counts control the amount of missing input, not its difficulty. Record the length of missing intervals and which body regions remain visible. Give each mask family the same training-only tuning opportunity, with several feasible coverage levels rather than assuming that either the current percentage or a video paper's high percentage is suitable. Keep source separation, paired random draws where applicable, architecture, teacher construction, readout procedure, and training exposure consistent. The present encoder retains masked positions in its token sequence, so hiding more tokens does not automatically reduce its computation as in a visible-token-only encoder.

Evaluate intact recordings and prespecified missing-region tests. For example, withhold a short ankle interval from every method and predict the original whole-clip laterality score from the remaining observations. That uses our existing kind of readout; reconstructing the ankle coordinates would require a separate decoder fitted on training videos. Report the direction of the left–right contrast as well as the magnitude of prediction errors, so systematically shrinking asymmetric scores toward zero does not look like an unqualified success. Choose both sides for masking without a permanent left/right preference; reflected-pair comparisons must transform the masks along with the landmarks.

For a test of genuinely missing observations, withhold them before preparing the evaluation input, and calculate normalization and interpolation using only permitted observations or an independently specified reference. The untouched recording supplies the reference score, not hidden information for input preparation. Masking an already prepared sequence remains a useful representation-sensitivity test, but supports a narrower claim. This distinction concerns the evaluation boundary; the full-input teacher remains a legitimate source of self-supervised training targets.

If structured masks improve prediction only when the matching region is missing, that would support a specific robustness benefit. If they also improve intact-video prediction over the initial encoder, the evidence for useful pretraining would be broader. If random subsets or soft preferences match the fixed gait set, the exact anatomical selection has no demonstrated special advantage. If all learned alternatives remain weak, the next explanation should concern the input, objective, or evaluation—not an assumed need for still more elaborate masks.

The most promising contribution is a controlled account of **which missing-information tasks preserve useful movement differences, and under what observation conditions**. The literature already supplies strong masking ideas; our opportunity is to determine how they behave when the outcome depends on fine left–right movement rather than broad action identity.

5. What recent research changes about the broader novelty claim

The literature supports the general motivation, while narrowing what would count as a distinct contribution. The following are primary research sources checked for this review; preprints are identified where proceedings publication was not verified.

Skeleton masking and feature prediction already have substantial precedent. Section 4 explains why anatomical or motion-aware masking alone is too broad a novelty claim. The ICCV 2025 paper *Towards Efficient General Feature Prediction in Masked Skeleton Modeling* also predicts local and global skeletal features. Our opportunity is a controlled explanation of when particular training targets preserve a useful movement quantity. [Skeletal feature prediction](#)

GaitJEPA already combines masked and future-latent prediction on walking silhouettes. Its authors report IJCB 2026 acceptance, with evaluations of person identification and sex classification. Those tasks

and inputs differ from our signed coordinate-based movement target, but the work rules out presenting the general combination of JEPA and gait as new. [GaitJEPA author repository](#)

Combining geometric consistency with useful features is an established research problem. The NeurIPS 2025 seq-JEPA paper separates representations serving invariant and transformation-sensitive tasks. Soft Equivariance Regularization, published at ICLR 2026, reports that deeper geometric supervision can improve equivariance scores while reducing classification performance, and studies intermediate-layer supervision instead. Our earlier separation between consistency and prediction should be situated within that literature. A stronger new result would identify which geometric relationship preserves signed body dynamics under the observation conditions in our data. [seq-JEPA](#), [Soft Equivariance Regularization](#)

What receives supervision matters. The 2026 V-JEPA 2.1 preprint studies supervision of visible as well as hidden tokens and multiple encoder layers. Its findings motivate a small comparison of training targets that preserve local movement, rather than assuming that predicting any hidden features will retain the information our task needs. Its results concern much larger image/video models and do not establish an expected gain for this dataset. [V-JEPA 2.1](#)

The August 2026 Human-JEPA preprint makes the evaluation of forecasting particularly relevant. It adapts a video JEPA to human imagery and reports that the future predictor adds almost no benefit beyond its encoder in an NTU-120 early-action evaluation using the first half of each clip. Improvements to the encoder and improvements from using predicted future features must therefore be measured separately. This motivates extending Notebook 10 beyond a probe of past features. Its RGB-video experiments are adjacent evidence, not results on our skeleton task. [Human-JEPA](#)

A world-model claim needs an evaluation of prediction over time. The 2026 LeWorldModel preprint evaluates action-conditioned latent prediction and control. It also reports limitations of its feature-distribution regularizer when observations have little diversity. Together with LeJEPA's proposed regularizer, this suggests a possible comparison, not a guaranteed remedy for weak gait prediction. Our recordings contain observational walking without controlled actions; successful forecasting would support a movement-dynamics result, while intervention or planning claims would need additional evidence. [LeWorldModel](#), [LeJEPA preprint](#)

Clinical transfer requires an appropriate movement representation and independent outcomes. The July 2026 GaitEncoder preprint evaluates gait features against clinical outcomes with participant-separated tests, using an evaluation encoder distinct from its final all-participant model. Its preparation flips left strides into a common right-stride convention with the stated aim of avoiding laterality encoding. Those processed inputs would need reconsideration for our signed target; stride-normalized kinematics also differ from our timestamped image-space landmarks. Its strongest lesson here concerns study design, not a ready-to-use external benchmark. [GaitEncoder preprint](#), [author repository](#)

6. The most productive research directions

Direction A: Explain and recover access to movement information

This is the first priority because the complete masking result shows weak prediction even before asking which anatomical targets are best. It connects Notebook 07's measurement discrepancy with Notebook 08's controlled comparison.

Start with a small set of separately interpretable changes: retain timing more faithfully during preparation; preserve movement order in the feature summary; and select regression regularization on training-source validation groups. Keep the compared clips and outer test groups fixed. Check the untrained and trained encoders using the same procedure, so an improved regression model cannot be mistaken for improved pretraining.

For example, if training-only penalty selection improves trained and untrained features equally, the main improvement concerns the readout. If preserving time substantially improves a direct movement baseline but leaves learned features unchanged, the training representation still deserves investigation. If a motion-sensitive feature summary reveals a repeatable learned-over-initial advantage, that would support the more specific explanation that the old summary obscured useful learned content.

In a follow-up run, predefine checkpoints and compare prediction from the actively optimized encoder and the slowly updated target encoder on training-source validation groups. The retained masking scores use the latter, and separate predictive learning curves have not yet been saved. These comparisons could help select a defensible training budget without using outer-test scores for selection. Neither the update schedule nor decreasing training loss proves why prediction is weak.

After establishing an informative evaluation, use Section 4's staged masking comparison to separate the effect of anatomical selection from movement weighting and the shape of missing observations. A gait-specific advantage that persists against matched random sets and structured alternatives would be more informative than the present two-choice comparison. Add visible-token or intermediate-layer supervision separately if the diagnostics justify it.

For a concrete test of observation loss, hide a short ankle segment and ask whether the remaining hip, knee, and opposite-leg observations help predict the laterality score calculated from the unaltered recording. Follow Section 4's preparation safeguards and compare with randomly placed gaps of the same size, retaining the same clips and reference targets. Such a result could show when learned body relationships help beyond direct measurements of the available coordinates. The reference would still be pose-derived, so clinical accuracy would remain a separate question.

The potential paper would explain how a seemingly sensible pretraining comparison changes when movement information and the prediction procedure are controlled. Its novelty would depend on demonstrating a generalizable mechanism or remedy; adjusting one regression parameter alone would be a modest contribution.

Direction B: Test whether reflection changes the features in a predictable way

Our current feature test assumes that each numerical feature should stay unchanged after exchanging anatomical landmarks. A learned encoder could instead express reflection through a predictable change in those feature values.

For illustration, suppose two features encode “overall movement” and “left-minus-right movement.” Reflection should preserve the first and negate the second. Comparing both coordinates as if neither should change would penalize a useful representation. This example does not show that our encoders have learned such features; it identifies an alternative explanation worth testing.

A low-compute experiment could freeze each encoder and learn one shared transformation of feature values from original/reflected training pairs, after aligning anatomical landmarks. Evaluate that transformation on held-out videos. Applying it twice should return to the starting point. Requiring it also to preserve lengths is an additional modeling restriction that prevents shrinking all features to obtain better agreement; this restriction is not guaranteed by reflecting the input. Keep the original unchanged-feature score, and compare real pairs with mismatched pairs and trained encoders with their initial references. Failure of this restricted transformation would leave more general reflection responses untested.

If a transformation fitted on training sources explains held-out reflection behavior, it would support a response within the features that the original test does not capture. This expands the assumed reflection rule; simply rotating the coordinate system cannot turn a nontrivial response into an unchanged one. To support a useful-learning claim, relate that structure to signed prediction or forecasting, with independent evaluation. Failure to generalize, similar performance on mismatched pairs, or an equally strong result

before training would weaken the learning explanation. Lower alignment error alone would not establish improved prediction.

This direction is a good match for NeurReps because it turns a concrete experimental ambiguity into a question about learned representations. Related equivariance methods already exist; the contribution would be the explanation and its predictive consequence in noisy body observations.

Direction C: Make geometric structure improve observable future prediction

This has the greatest potential for a paper about movement world models, with substantially more empirical work required. First complete Notebook 10's real-data comparison. Then evaluate the JEPA predictor's future features themselves. Fit a decoder—a model converting features into movement estimates—using the target encoder's features of observed future windows from training videos only, and freeze it for evaluation. On held-out videos, compare decoding the observed future features with decoding the predicted future features. If the first fails, the decoder or feature representation is already inadequate; if only the second fails, future-feature prediction is the more direct suspect. Also compare with regression directly from past coordinates and past encoder features.

For example, after observing through 0.8 seconds, ask whether a predicted future feature vector can recover ankle displacement at 1.3 seconds. Repeat on the reflected input and check whether the predicted movement transforms appropriately. A model that improves its reflection score but cannot predict the displacement has not demonstrated the desired benefit.

Retain the direct past-pose, initial-encoder, and mismatched-future controls. To test the value of temporal order, disrupt the association between earlier positions and their timestamps while keeping the last two observations fixed, preserving the immediate position and velocity references. Merely reordering observations together with their timestamps would leave their temporal information intact. Use common evaluable landmarks and clips when comparing methods, and report changes in coverage across horizons. If correct futures or correct temporal associations do not help, apparent success may rely mainly on static pose or smoothness.

Once one-step latent prediction is useful, a further implementation could feed predicted states into subsequent predictions. The current predictor accepts pose windows, so this would require a compatible feature-to-feature transition or an explicitly tested decode-and-re-encode procedure. Asking about several horizons from the same observed prefix does not yet implement repeated prediction. Compare reflection penalties at intermediate versus final features as a separate experiment. Synthetic occlusion and pose perturbations can test robustness to observation errors, but they are not physical interventions on a person.

A convincing result would show where geometric structure improves future movement prediction and where it costs accuracy. An observational forecasting model could support a focused world-model contribution without claiming action planning, clinical decision support, or a complete physiological simulator.

When expanding external evaluation is worthwhile

Notebook 06 is worthwhile if it leads to an actual independent evaluation. Additional readiness checks alone add little scientific evidence. Select the outcome and dataset first, verify coordinate and timing compatibility, and establish simple baselines before transferring the selected method.

An external set could test generalization to new people where participant separation from development data can be verified and those people are excluded from training and model selection. Any unresolved overlap must be reported. An independently measured movement quantity would test more than reproducing the same coordinate formula. A clinical rating would support a clinical question only if the data,

reference, and study design justify that interpretation. These are different goals, and one external dataset need not serve all of them. The existing [external evaluation assessment](#) discusses candidate routes and their unresolved requirements.

Because the current recordings have already shaped our hypotheses, further comparisons on them should be described as development evidence. An independently specified evaluation would provide a stronger test of the selected explanation.

7. How close is this to a strong workshop paper?

The completed grid improves the evidence base substantially. We can now report a full controlled masking comparison instead of a partial run. The central anatomical-benefit hypothesis remains unsupported, however, and its explanation is unresolved. A credible negative-result paper is possible; a claim of improved JEPA learning requires further evidence.

The following scores are editorial judgments on a five-point scale, not estimated acceptance probabilities. Five means a strong position on the stated dimension; one means substantial missing evidence or poor fit.

- **Importance of the question: 4/5.** Distinguishing useful movement learning from satisfying a training or consistency objective matters for body representation learning.
- **Experimental control: 4/5.** The complete source-separated, paired grid and matched hidden-token counts are strong. Regression selection and anatomy-specific controls need further work.
- **Evidence for the current bounded conclusion: 4/5.** The full grid supports reporting that this procedure fails to establish the proposed masking benefit. It does not support a universal negative claim.
- **Explanation of the result: 2/5.** Input preparation, temporal summaries, regression regularization, and training dynamics remain competing explanations.
- **Novel contribution beyond related work: 2/5 today.** The ingredients and general consistency/usefulness distinction have precedent. A specific mechanism, convincing correction, or new predictive result would strengthen novelty.
- **External generalization: 1/5.** No independent participant-level or independent-outcome evaluation has been completed.

For the proposed directions, the scores separate evidence already available from potential after successful experiments. “Evidence now” rates the support for investigating the question, not evidence that the proposed remedy works. Feasibility concerns completing a focused comparison with the existing data and tools; workshop potential assumes a clear, well-supported outcome.

Research direction	Novelty potential	Evidence now	Feasibility	Workshop potential
A. Recover movement information	3/5	3/5	5/5	4/5
B. Explain the feature response to reflection	3/5	2/5	4/5	4/5
C. Improve future movement prediction	4/5	1/5	2/5	5/5

Direction A has the best near-term return because it uses completed experiments to investigate a concrete failure. Its strongest paper would explain more than a dataset-specific tuning issue. Direction B is an economical, focused geometry question with a clear way to reject its explanation, although demonstrating

useful consequences is essential. Direction C offers the strongest temporal-model contribution, but its high potential score is conditional: no real-data forecasting benefit has yet been demonstrated.

Which workshop audience is most appropriate?

Foundation Models for the Brain and Body: topic fit 5/5. Its call includes movement, video-derived pose, and evaluation of pretraining. Direction A addresses whether a body encoder learns useful information, while Direction C could add temporal evidence. A broad foundation-model capability would still require transfer across tasks or datasets. The posted paper deadline was 5 September AoE and has passed. Its separate interactive-demo call remains open until 19 September AoE, but a demonstration would be a different submission from a research paper. [Paper call](#), [demo call](#)

NeurReps: topic fit 5/5; track choice matters. Direction B directly concerns the geometry of neural representations. The four-page, non-archival extended-abstract track is the best format fit for the present controlled findings and developing explanation. The call advertises a nine-page archival proceedings track for more developed work. Its **Findings Track receives 2/5 for our current work**: it emphasizes high-impact collaborations between experimentalists and theorists, including biological discoveries with substantial geometric insight. It uses editorial review and has no page limit. “Findings” here does not mean an easier route for any negative experimental result. The posted deadline was 24 August AoE; no extension was verified. [Official track descriptions](#)

Physical World AI: topic fit 4/5. Articulated geometry and evaluation of learned physical representations are relevant, with Direction C offering a stronger temporal connection if completed. The present work does not evaluate contact, materials, or multimodal sensor fusion. The call advertises an archival deadline of 9 September and a non-archival window of 29 September–29 October, with eight-page papers or four-page extended abstracts. That later window is the most practical posted paper route among the previously considered venues for developing a focused result. [Official call](#)

Foundation Models for Temporal Systems (FMTS): topic fit 4/5 for Directions A and C. This additional workshop explicitly includes irregular sampling, sparse observations, leakage-aware evaluation, and temporal consistency. It accepts four-page non-archival submissions, including preliminary and negative findings, until 15 September AoE. A completed analysis of movement timing or forecasting would fit; a general roadmap alone would not establish a temporal-learning contribution. [Official call](#)

Embodied Spatial Reasoning: topic fit 3/5. Body geometry and movement prediction connect to spatial and temporal reasoning, but we do not yet study an interacting agent, surrounding objects, or spatial memory. The workshop welcomes preliminary and negative results; its posted 5 September AoE paper deadline has passed. [Official call](#)

GenAI4Health: topic fit 2/5. Careful evaluation is relevant to clinical trust, but a coordinate-derived gait result does not supply the missing medical generative application or clinical benefit. Its 9 September AoE deadline is close, and the evidence should determine the claim rather than the available deadline. [Official call](#)

Med-Reasoner: topic fit 1/5. The current pose-only model does not address the workshop's medical vision-language reasoning problem. Its posted 5 September AoE deadline has passed. A future submission would require a genuine clinical reasoning task and evidence about that reasoning, not simply health-related input data. [Official call](#)

NeurReps and PhysWorldAI advertise archival options, while central NeurIPS guidance describes workshop papers as non-archival. This unresolved policy difference should be checked with organizers before choosing an archival route or relying on its compatibility with another submission. [NeurIPS workshop guidance](#)

Why IAAI remains a different research trajectory

IAAI-27 has an application focus, giving the current work a fit score of **1/5**. Its emerging-applications track seeks pilot or early-deployment evidence and a credible path to use; its deployed-applications track requires production use and measured benefits. The current notebooks document neither. The clearest route would include an intended user and decision, an end-to-end system, evaluation against the existing workflow, a real user pilot, and evidence about failures and operation. These would be new application work, not a refinement of the present representation-learning results. [IAAI-27 call](#)

The posted IAAI deadline is 8 September AoE. Its restrictions on overlapping submissions explicitly exempt limited-audience workshops without archival proceedings; this corrects an overly broad reading of the policy. Each destination's own rules still need checking. The near deadline does not resolve the missing application evidence. [Submission policy](#)

8. A practical sequence toward the next paper

1. **Preserve and explain the complete masking result.** Report the full source coverage, initial-encoder comparison, and uncertainty. Inspect large-error videos and sensitivity to random training choices without removing unfavorable cases.
2. **Resolve the smallest plausible explanations first.** Compare training-only regression selection, temporal preparation, and motion-sensitive summaries in controlled steps. In parallel, test the frozen-feature reflection transformation from Direction B. Record which explanation each experiment could support or weaken.
3. **Choose one main contribution.** If the information-recovery experiments identify a repeatable mechanism, develop a focused body-representation paper. If the learned reflection rule explains feature behavior with a useful consequence, develop the geometry paper. Pursue the larger forecasting study when its simple real-data controls establish a credible starting point.
4. **Confirm the selected result independently.** Freeze the principal choices before the external evaluation and make the intended generalization claim explicit. Keep clinical interpretation proportional to the available reference measurements.
5. **Write around the answer, including an unfavorable one.** A workshop paper should make one question, its decisive comparison, and its implications easy to follow. Avoid presenting the entire notebook roadmap as demonstrated novelty. Finalize the applicable ethics, data-use, and submission-policy documentation separately from the scientific assessment.

The project has progressed from a sensible anatomical training choice to a more demanding question about useful learning. The completed masking result gives us a concrete problem to explain, and the new notebooks provide ways to distinguish possible explanations. The strongest next paper will come from resolving one of those questions with a clear comparison and an observable movement outcome.

9. Copy-ready implementation prompt for the next notebook suite

The following prompt specifies a future implementation task. The notebook names below are proposed deliverables, not notebooks created by this document update. Copy the text from “Role and objective” through “Definition of done” when ready to implement the extension.

Role and objective

Act as an AI/ML researcher and research software engineer specializing in skeletal representation learning and JEPA. Extend the tutorial notebooks in **neurips-laterality** to compare masking policies, the encoders trained with them, and the resulting predictor performance. The central question is which missing-information tasks help preserve useful movement differences on recordings excluded from training.

Begin by reading the applicable project instructions, `docs/TUTORIAL.md`, notebooks 00–10, their editable tutorial sources, the relevant training and evaluation helpers, and the retained real-data results. Check the primary papers cited in Section 4 before implementing a named method. Follow the user's plain-research-writing preferences. Use subagents for independent implementation, testing, and adversarial review, with explicit file ownership to avoid conflicting edits.

Implement and verify the notebook suite, but do not launch the full real-data training grid automatically. Default execution must use inexpensive synthetic examples. Provide a clearly labeled, explicitly enabled path for real-data experiments and display the complete planned workload before training starts. Never invent performance values or substitute synthetic examples for missing empirical results.

Preserve the completed research

Keep notebooks 00–10, their tutorial sources, `laterality/`, the existing protocol, saved results, checkpoints, and executed copies unchanged. In particular, do not extend `laterality_extensions/masked_learning.py` in place: modifying that implementation can invalidate compatibility checks for the completed masking comparison. Put new policies and runners in new modules, reusing unchanged components where appropriate.

The next unused notebook number is 11 at the time of this prompt; check again before creating files. Follow the existing editable-source convention with `tutorials/research_11.py` and subsequent numbered sources. Extend `scripts/build_research_notebooks.py` carefully and generate only the selected new notebooks with `--only`; its default behavior regenerates every notebook in its mapping. Preserve output-free source notebooks and save executed teaching copies separately. Add focused test files without weakening existing tests, and verify that protected files remain unchanged.

Create four connected tutorials

1. **11_masking_patterns_and_coverage.ipynb**. Explain the policies and implement their construction before meaningful model training. Show what is hidden and what remains observable, using readable vector graphics. Demonstrate eligibility, valid-token counts, temporal spans, anatomical coverage, and impossible-budget cases. Clearly separate naturally missing observations from deliberate training masks.
2. **12_controlled_masking_pretraining.ipynb**. Implement a shared training runner with matched initial states, source sampling, augmentations, encoder, predictor, teacher construction, and optimizer exposure. Separate the anatomical-selection comparison from the motion and mask-shape comparison. Include a tiny synthetic execution and an explicit real-data configuration, with safe saving and reuse of completed compatible runs.
3. **13_masking_encoder_and_predictor_evaluation.ipynb**. Evaluate both the pretraining predictor and the usefulness of frozen encoder features. Include initial encoders, direct-pose summaries, and training-mean controls. Aggregate source-held-out predictions correctly, inspect declared on-line/teacher checkpoints, and evaluate prespecified missing-observation conditions. Explain why a lower feature-prediction loss may coexist with weak movement prediction.
4. **14_future_features_and_movement_prediction.ipynb**. Build on the question in Notebook 10 without rewriting it. Separate regression from past features from decoding the JEPa predictor's future features. Provide the training-only decoder control described below, simple forecasting references, and tests excluding future information from the input. Keep twelve- versus all-landmark input as a separate configurable comparison with the same future landmark endpoints. Its implementation and synthetic tests are required; real training requires explicit enablement.

Each notebook should introduce its question and dependencies, explain consequential design choices before the code, and use small code cells that expose the method. Move repetitive machinery into

readable helpers. Explain each displayed result in terms of what it supports, what it leaves unresolved, and what outcome would weaken the hypothesis. Avoid unexplained condition nicknames, long decimal strings, local run identifiers in the narrative, and claims of neurological validity unsupported by the experiment.

Separate the experimental questions

Before implementing the training grid, write a compact experiment specification distinguishing:

- **Landmark eligibility:** gait-only targets, all-landmark targets, fixed random subsets, and a soft gait preference, with scattered mask geometry held constant.
- **Mask shape and movement weighting:** scattered, motion-weighted, complete-trajectory, and connected-region masks under the same declared landmark eligibility.
- **Time prediction:** an interior missing interval with observations on both sides versus a future interval with observations only before it.
- **Other model choices:** input landmarks, regularizer pooling, temporal feature summaries, predictor capacity, and target/loss changes. These belong in separate comparisons.

For the first masked-learning comparisons, retain the existing 33-landmark input, twelve-landmark pooling of unmasked regularizer views, five-pair laterality summary, and coordinate-derived target. Describe these remaining anatomical choices openly. Do not call the all-landmark masking condition free of all anatomical guidance. Do not describe the extension as an exact reproduction of the original paper's training schedule.

Define the masking policies precisely

Implement a common policy interface that receives the permitted observations, token-validity mask, budget specification, and a dedicated random generator. It should return the deliberate mask, unique target positions, remaining-context counts, and a coverage summary. It must not receive the laterality label or clinical-condition annotations.

- **Gait-only scattered masking:** preserve the existing twelve-landmark eligible set and redraw joint-time targets at every update.
- **All-landmark random masking:** sample without replacement from every valid joint-time token, with equal initial sampling weight.
- **Fixed random twelve-landmark sets:** preselect several sets with separate recorded selection seeds, reuse them across folds and training seeds, and report them as a declared group. Random individual landmarks and random bilateral pairs are different controls; name them accordingly. Do not select a favorable subset after seeing prediction results.
- **Soft gait preference:** keep every valid token eligible. A starting rule is an equal mixture of uniform all-landmark probabilities and uniform gait-landmark probabilities, followed by weighted sampling without replacement. Specify the fallback when no gait candidates are valid. This mixture does not guarantee that exactly half the sampled targets come from each component, and its weight is a proposed comparison setting rather than a validated optimum.
- **Motion-weighted masking:** make greater reliable movement increase the probability of being hidden, consistent with MAMP and S-JEPA. Specify how movement is computed, its timing units, normalization, treatment of missing transitions, and control of isolated jumps. Retain a uniform sampling component and define behavior for stationary or unusable motion. Keep any motion-target reconstruction objective separate from this sampler.
- **Whole-joint trajectories:** hide selected landmarks throughout the entire input window wherever their observations are valid. Do not shorten the trajectory to satisfy a count while retaining the whole-trajectory label.

- **Connected-region intervals:** use explicit anatomical connections to select a connected region and a consecutive time interval. Ensure the region definitions collectively cover the declared candidate pool. Array-index adjacency is not an anatomical graph.
- **Interior temporal gaps:** hide every valid landmark in a contiguous interior interval while retaining observations before and after it. Evaluate this as completion; implement future-only masks separately in the forecasting tutorial.

Complementary views, learned selectors, broad mixtures of mask families, and visible-token supervision should remain optional follow-ups. Do not add them to the primary grid merely because the interface can support them.

Make token-budget feasibility explicit

Preserve the current four-step tokenization and validity definition unless a separately named experiment changes them. Count valid input tokens, deliberately hidden valid tokens, remaining valid context, and the fraction hidden. Missing measurements cannot become supervised targets, and overlapping blocks cannot count a position more than once.

Match realized hidden counts exactly for compatible comparisons. Require at least one valid context token. When a new policy lowers the shared feasible count, rerun its reference conditions at that count. Record the actual interval lengths, landmark coverage, and body-side coverage rather than relying on a policy name or requested percentage.

Check counts within each batch as well as between policies. The existing prediction path flattens selected targets and reshapes them by batch, assuming the same count for every example. If structured masks and missingness produce unequal counts, either enforce a feasible equal-count invariant or implement a new dense or packed target path that preserves sample and token identities. Do not reuse the old reshape on unequal counts. Define loss weighting explicitly—for example, average over each example's hidden targets before averaging examples—and test that targets cannot be mixed between clips.

Do not require an impossible combination of exact counts and intact mask shapes. With sixteen time blocks and 33 fully valid landmarks, whole trajectories contribute multiples of sixteen and whole time blocks contribute multiples of 33. Their only common positive count is the whole input, which leaves no context. Use separate count-matched scattered references, a declared coverage-range analysis, or separate experiments for these families. Never trim a structured mask or add isolated cells silently to achieve an arbitrary count, and never describe approximate matching as exact.

Control training, randomness, and model selection

Use the established five source-video outer groups and five training seeds for the declared full comparison. Every clip from an outer-test video must remain excluded from encoder training, selection of policies or checkpoints, and fitting the movement readout. Repeated seeds represent repeated training choices on the same recordings, not new independent participants.

Use separate reproducible random streams for initialization, source sampling, geometric views, policy-specific mask draws, random-subset selection, and evaluation corruption. An adaptive policy's additional random draws must not change which training clips its reference model receives. Pair compatible initial weights, source schedules, augmentations, and update counts; save evidence that these controls hold. The realized mask itself should differ when mask shape is the intended comparison.

Keep architecture, predictor capacity, teacher update rule, learning-rate schedule, regularizer, batching, training exposure, and readout procedure fixed within a masking-only comparison. If a later experiment changes predictor capacity or encoder dimensions, retain a matched initial control for each architecture

and state which parameters can still be paired. Report compute or runtime as well as update counts; equal updates need not imply equal cost.

Extract and display the complete real-data recipe from the saved Notebook 08 comparisons as the starting reference. Do not inherit a small synthetic demonstration's settings for real training. Preserve its 1,200-update budget for a direct extension, or document a different budget selected through training-only evidence and apply it consistently to the compared methods. List all model, optimizer, schedule, teacher, masking, and regularization settings before starting a run.

Predeclare a small primary comparison and a limited training-source validation procedure. Use inner source-separated validation for regression penalties and any mask, budget, or checkpoint choices. Do not use the outer test results to select the best policy or early stopping point. If an inner validation result is intended to assess an entire pretraining recipe, also exclude those validation sources from that candidate's encoder training. Distinguish this from tuning only a supervised readout on representations already pretrained within the outer training set.

Check the current evaluation's ability to detect useful information before expanding training. In a synthetic test with a known movement signal, demonstrate that an appropriate simple readout can recover it; this verifies the evaluation pipeline without implying a real-data gain. Treat subsequent choices informed by the already inspected GAVD results as development decisions, with independent evaluation needed for stronger confirmation.

Evaluate three different kinds of prediction

First, measure the JEPA predictor's behavior. Evaluate hidden-feature prediction on a fixed, declared bank of evaluation masks and include appropriate initial and mismatched-target controls. Preserve its normal pathway: online/context encoder to predictor, with the teacher supplying targets. Record feature variation for targets and predictions, feature norms, effective rank, and a clearly defined normalized error with its denominator. Compare online and teacher features through separately fitted frozen readouts at prespecified checkpoints, using training-source validation for any selection. Substituting the teacher into the predictor's input pathway would be a separate experiment.

Errors against each arm's own learned teacher are not a common measurement scale: a model can make prediction easier by producing less variable targets. Do not rank useful learning by raw training loss or assume that normalization fully resolves this issue. A common frozen teacher could support a separate comparison trained against that shared target. Do not score existing predictors directly against a different teacher's vectors without an explicitly fitted and controlled alignment; their latent coordinates are not automatically compatible. The between-method usefulness claim should rest on a shared observable movement outcome.

Second, evaluate frozen-feature movement prediction. Fit the same training-only imputation, scaling, and ridge-selection procedure for every representation. Retain the matched initial encoder, direct-pose summaries, and a mean target estimated from training sources. Keep online and teacher results distinct, and use a common, declared primary endpoint rather than selecting whichever metric favors a method.

Use source-balanced R^2 as the primary laterality score and mean absolute error as a complementary error measure. Report sign behavior where useful, while explaining instability for targets close to zero. Within each seed, pool all out-of-fold predictions before computing the metric; then summarize seed-specific scores. Do not average fold R^2 values or silently replace the analysis with an ensemble of seed-averaged predictions. Include evaluated clip and source counts for every table.

Third, evaluate future-feature prediction in Notebook 14. After fitting the forecasting model on training sources, freeze it and fit a decoder from observed future-teacher features to future coordinates

using training videos only. Freeze that decoder and evaluate it on both observed and predicted future features from held-out videos. The observed-future condition checks whether the features and decoder can express the endpoint; it is a diagnostic with access to the future, not a deployable forecasting baseline. Demonstrate this path synthetically first and leave real-data results absent until their experiment runs.

Compare with regression directly from past coordinates, regression from past encoder features, an initial encoder, persistence, recent-velocity continuation, and an encoder trained against mismatched futures. Sample mismatched futures from other training sources at the same horizon, balancing sources before clips. Keep horizons, prediction units, and evaluated observations identical across the relevant methods. Report actual observation times when sampling does not land exactly on a requested horizon.

Notebook 10's existing past-feature regression must retain its current interpretation. It does not decode the JEPA predictor's future feature vector. Do not call multiple forecasts from one fixed prefix a recursive rollout; recurrent prediction requires a compatible latent transition or an explicitly tested decode-and-re-encode procedure.

Test missing observations without exposing withheld information

Create a shared, prespecified bank of evaluation corruptions, including scattered gaps and connected missing regions on both sides of the body. Every compared model must receive the same corrupted observations. Keep the laterality score from the unaltered recording as the reference; predict that score from the remaining observations. The unaltered recording may already have missing measurements. Actual coordinate reconstruction requires its own decoder and metrics.

For a claim about genuinely missing data, remove the selected raw observations before interpolation and normalization, using only permitted observations or independently specified reference geometry to prepare the input. A test masking already prepared coordinates should be identified separately as representation sensitivity. Report corruption feasibility, unavailable predictions, and changes in coverage; do not silently discard difficult cases or make each method's test set more favorable.

Ordinary full-input teacher targets during self-supervised training remain legitimate. For past-only forecasting, however, input preparation, context-derived motion scores, and any adaptive selection must use only the observed prefix. The teacher can use the declared future window for training targets. Changing future values must not change the input available to the context encoder or its deterministic predictions.

Require tests before interpreting real results

Add focused tests that establish all of the following:

- Identical seeds and inputs reproduce masks; outputs contain no duplicate targets, no invalid targets, and at least one valid context token.
- Fixed subsets remain fixed while masks are redrawn; soft preference retains non-gait eligibility; motion-weighted sampling hides higher-motion tokens more often in a repeated-sampling test, with valid stationary and noisy-input behavior.
- Complete trajectories, connected regions, and temporal gaps satisfy their definitions. Infeasible budgets fail clearly or follow an explicitly documented alternative.
- Reflection exchanges coordinate signs, anatomical identities, validity, and mask locations consistently. Both anatomical sides can be selected.
- Withheld content does not enter the relevant student or evaluation pathway. Hold the selected mask fixed when perturbing hidden values to test direct content isolation. Motion-based selection during ordinary self-supervised training can legitimately respond to changes in the complete permitted observation; future-informed selection remains forbidden in a past-only input. Changing labels cannot alter label-blind pretraining when splits and random streams are fixed.

- Teacher parameters receive no gradients; online and teacher checkpoints are stored and identified separately.
- Fitting functions cannot access outer-test sources. Altering outer-test values leaves trained parameters and selected hyperparameters unchanged.
- Perturbing future coordinates, visibility, and timestamps that remain after the boundary leaves prepared context and past-only predictions unchanged. Moving an observation across that boundary is a different test.
- Constant or nearly constant features trigger an explicit diagnostic rather than appearing to demonstrate successful learning.
- Aggregation rejects duplicate predictions, cross-fold source overlap, incomplete declared coverage, and incompatible paired comparisons. A partial run can be summarized only with its actual scope clearly labeled.
- Fresh-kernel execution completes the synthetic tutorials and renders their figures without relying on hidden notebook state.

Tests should verify the implementation and scientific controls. They must not require a proposed mask to outperform a baseline; an unfavorable empirical result is a valid outcome.

Retain results and quantify uncertainty carefully

Use separate output locations for the new experiments and refuse to overwrite completed results. Save configurations, data and split references, subset definitions, budgets, realized coverage, source schedules, loss components, declared checkpoints, and per-clip predictions with their source, fold, seed, condition, target, and prediction.

Only reuse a completed run when its data, split, policy, model, objective, schedules, random seeds, and implementation are compatible. Corrupted, incomplete, duplicated, or mismatched saved results must fail clearly. Save atomically. If resuming training is supported, retain and restore optimizer, teacher, model buffers such as target centers, every sampler's random state, schedules, and global update position. Test agreement with an uninterrupted run within declared numerical tolerances; otherwise label interrupted runs incomplete.

For paired uncertainty, resample complete source videos while keeping their clips, conditions, and seed predictions together. State that these intervals condition on the fitted models and do not represent full retraining uncertainty. Report variability across seeds separately. Keep full precision in machine-readable results and use sensible rounded values in the tutorial.

Before any real run, display the selected conditions, folds, seeds, updates, approximate workload, output locations, and whether the request is a pilot or the complete declared comparison. Do not launch a large Cartesian product of all masks, ratios, models, and endpoints by default.

Definition of done

The implementation is complete when the new source notebooks validate and execute their synthetic examples in fresh kernels; mask and evaluation tests pass; protected research files are unchanged; and the explicit real-data path can validate its inputs and display its workload without starting training. The documentation must distinguish implemented methods, software demonstrations, completed empirical comparisons, and analyses still awaiting results.

Use independent reviewers to check mask definitions and budgets, source separation and evaluation, and the clarity of the explanations. Fix substantive findings and rerun the affected checks. In the handoff, state which notebooks and helpers were added, what was executed, what remains unrun, and which

scientific conclusions the retained evidence supports. Update the tutorial and its PDF when new results exist, without inventing gains or treating a useful anatomical prior as established neurological validity.

End of implementation prompt.

Evidence and figure reproduction

This overview combines the completed results in notebooks 00–05, the diagnostic procedures in 07, the full saved real-data masking comparisons associated with 08, and the retained synthetic executions of 09–10. Source notebooks may be output-free; absence of inline output was not treated as absence of a saved experiment. No real-data symmetry-training or forecasting result was found in the reviewed material.

The masking table and figure use one pooled held-out score per seed, followed by averaging those scores. The exploratory uncertainty calculation resamples the 93 source videos 20,000 times, keeping each video's clips and all paired predictions together; it conditions on the already fitted models. It is a new analysis of retained predictions, not a confidence interval previously stored by the notebook or an estimate of full retraining uncertainty.

The [result-figure and summary generator](#) reproduces the empirical masking summary and Figure 1 from saved predictions. Its [numerical summary](#) retains the unrounded values; prose, tables, and figure labels use precision appropriate for interpretation. The separate [mask-pattern illustration generator](#) draws Figure 2 and verifies that its four hypothetical panels hide equal numbers of cells; it does not run an experiment. Run the [PDF build script](#) after editing this Markdown source to regenerate [TUTORIAL.pdf](#). Existing notebooks, training code, and empirical artifacts were left unchanged by this document update.